Quiz: Retrieval-Augmented Generation (RAG)

Name: _____Ankit Raj_____
Date: _____16/03/2026_____
Score: _____ / 20

Section A: Multiple Choice Questions (1 mark each)

1. What is the main purpose of Retrieval-Augmented Generation (RAG)?
A. To compress large datasets
B. To combine document retrieval with language generation
C. To train a neural network from scratch
D. To replace vector databases

---

2. In a RAG system, what is the role of a vector database?
A. Store trained models
B. Store embeddings for similarity search
C. Generate answers
D. Train transformers

---

3. What does “chunking” refer to in RAG pipelines?
A. Dividing a document into smaller sections
B. Encrypting documents
C. Compressing text data
D. Converting text into images

---

4. Which method is commonly used to convert text chunks into vectors?
A. Tokenization

B. Embeddings
C. Sorting
D. Indexing

---

5. Which component generates the final response in a RAG system?
A. Vector database
B. Embedding model
C. Large Language Model
D. Data loader

Section B: True or False (1 mark each)

6. RAG helps LLMs use external knowledge sources.
True / False

7. Chunking large documents improves retrieval accuracy.
True / False

8. Context injection means adding retrieved information into the prompt.
True / False

9. In RAG, the LLM directly searches PDFs without embeddings.
True / False

10. Evaluating a RAG system only requires checking model accuracy.
True / False

Section C: Short Answer (2 marks each)

11. Write the basic workflow of a RAG system.

Documents → Chunking → Embeddings → Vector Database → Retrieval → Context + LLM → Generated Answer.

---

12. Why is chunk size important in RAG pipelines?

Chunk size is important in Retrieval-Augmented Generation (RAG) pipelines because it affects how well the system retrieves relevant information from documents.

- Better Retrieval Accuracy
- Improved Embedding Quality
- Efficient Similarity Search
- Maintaining Context for the LLM

---

13. What is context injection and why is it useful?

Context injection is the process of adding relevant retrieved information (documents or chunks) into the prompt of a language model before generating a response.
It is useful because it provides accurate and relevant knowledge to the model, reduces hallucinations, and helps the model generate more correct and context-aware answers.

---

14. Name two evaluation metrics or considerations for RAG systems.

 Retrieval Accuracy – Measures how well the system retrieves relevant documents.

 Answer Relevance / Response Quality – Evaluates whether the generated answer correctly uses the retrieved information.

---

15. Give one real-world application of RAG.

A legal research assistant that retrieves relevant laws or case documents and generates accurate answers for lawyers.

✅ **Total Marks: 20**

-